

MARQUETTE COUNTY AP, MI, USA

WMO: 727430

Lat: 46.5310N

Lon: 87.5490W

Elev: 1415

StdP: 13.96

Time Zone: -5.00 (NAE)

Period: 90-99

WBAN: 94850

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.1	-8.9	-23.9	1.5	-14.0	-17.3	2.3	-7.3	24.8	14.9	22.2	13.3	6.3	270	0.686

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.5	83.8	68.0	80.7	65.7	77.3	64.1	70.9	80.4	68.6	76.5	66.8	74.1	11.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.8	107.8	75.1	65.7	99.9	73.5	63.8	93.4	71.6	35.5	81.2	33.7	76.5	32.1	74.3	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.8	18.8	17.6	-23.4	90.9	6.8	3.2	-28.3	93.3	-32.2	95.2	-36.0	97.0	-40.9	99.3
			-23.6	74.9	6.7	2.8	-28.5	76.9	-32.4	78.5	-36.1	80.1	-41.0	82.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.2	13.4	16.4	23.9	36.0	49.1	59.6	62.6	61.8	53.7	42.9	28.7	20.5
	DBStd	20.01	11.29	12.42	11.47	10.52	9.88	8.92	6.60	7.05	8.69	8.35	8.57	10.95
	HDD50	5360	1136	941	811	443	142	19	1	2	57	253	641	914
	HDD65	9606	1601	1361	1275	874	505	206	130	148	352	686	1089	1379
	CDD50	1400	0	0	1	21	113	307	390	366	168	33	1	0
	CDD65	171	0	0	0	3	10	43	54	48	13	0	0	0
	CDH74	1515	0	0	0	28	138	406	467	368	98	10	0	0
	CDH80	331	0	0	0	9	24	99	108	70	19	2	0	0
Wind	WSAvg	9.0	9.7	9.6	9.1	9.3	8.8	7.8	8.0	7.8	8.9	9.8	9.9	9.6
Precipitation	PrecAvg	36.0	2.4	1.7	2.9	2.6	3.1	3.4	3.0	3.3	3.9	3.6	3.1	2.5
	PrecMax	51.6	6.6	3.7	6.1	6.6	7.9	12.2	5.6	8.6	7.6	7.6	8.3	6.9
	PrecMin	25.6	0.6	0.6	0.3	0.9	0.1	0.6	0.6	0.6	1.2	0.9	1.0	0.5
	PrecStd	6.3	1.4	0.9	1.7	1.3	1.7	2.2	1.5	1.6	1.8	1.6	1.7	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.3	47.0	57.4	77.1	83.2	88.7	88.1	86.2	82.2	73.5	57.1	46.4
		MCWB	34.1	37.9	47.9	60.3	62.6	73.7	70.5	68.7	66.9	59.4	54.4	39.3
	2%	DB	33.8	40.8	50.2	66.6	78.0	82.3	82.4	81.5	76.4	66.3	49.0	39.7
		MCWB	31.4	35.5	42.6	52.3	60.5	67.5	68.0	67.3	64.8	56.5	44.4	34.6
	5%	DB	31.1	36.6	45.4	59.1	72.9	78.7	79.3	78.6	71.9	61.4	44.4	36.2
		MCWB	29.6	32.4	39.3	48.7	57.9	64.4	65.6	65.6	62.6	54.5	40.4	33.0
	10%	DB	28.5	33.5	39.7	53.3	67.1	75.0	75.4	74.8	67.6	56.9	40.6	33.7
		MCWB	26.9	30.9	34.9	44.3	55.6	62.5	64.6	64.1	60.5	50.7	37.1	31.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.2	39.9	50.4	63.4	67.4	74.4	74.0	72.2	69.4	61.9	54.3	40.5
		MCDB	37.7	45.3	54.5	76.1	76.4	86.0	83.4	81.8	79.2	69.1	58.4	45.4
	2%	WB	31.9	35.3	45.1	54.9	62.9	70.4	70.8	69.5	66.3	58.6	45.8	35.8
		MCDB	33.4	39.3	49.1	61.4	73.1	80.0	79.6	76.6	73.4	64.2	47.7	39.1
	5%	WB	29.8	32.7	39.4	49.1	59.9	67.3	68.2	68.0	63.9	55.3	40.9	33.3
		MCDB	31.4	36.1	45.1	59.2	70.5	75.3	76.2	74.7	70.3	61.3	43.8	35.5
	10%	WB	27.3	30.6	34.9	44.2	56.3	64.5	66.0	66.1	61.6	50.5	37.3	31.3
		MCDB	28.5	33.2	39.7	52.8	65.5	71.7	73.4	72.5	66.5	56.7	40.0	33.6
Mean Daily Temperature Range	5% DB	MDBR	15.5	18.0	19.4	18.2	21.9	21.6	20.5	20.6	19.3	16.7	12.4	13.0
		MCDBR	15.9	20.3	23.4	29.6	28.3	25.6	25.4	23.6	24.1	25.2	17.3	14.9
		MCWBR	14.5	16.0	16.5	17.1	15.1	14.0	13.7	12.7	14.6	17.1	13.9	11.4
	5% WB	MCDBR	15.3	17.2	21.0	26.1	24.3	21.3	22.0	19.4	20.4	22.5	16.4	13.7
		MCWBR	14.2	14.0	15.5	16.3	14.3	13.2	13.1	11.9	13.7	16.8	14.4	11.2
Clear-Sky Solar Irradiance	taub	0.256	0.270	0.302	0.329	0.363	0.364	0.370	0.357	0.346	0.318	0.290	0.256	
	taud	2.433	2.381	2.387	2.416	2.392	2.412	2.423	2.457	2.490	2.546	2.539	2.486	
	Ebn at Noon	273	290	294	295	286	285	282	282	276	270	258	261	
	Edh at Noon	21	27	32	34	36	36	35	33	29	23	19	18	
All-Sky Solar Radiation	RadAvg	333	599	1058	1411	1725	1870	1908	1616	1192	705	373	259	
	RadStd	26	80	99	148	123	142	99	103	80	84	48	22	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page